

Sagittarius A* as a Metastable Entropy Modulator: Star Formation Near the Galactic Center in the UTAC Framework

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Repository: <https://github.com/GenesisAeon/Feldtheorie> (<https://github.com/GenesisAeon/Feldtheorie>)

Abstract

(see above)

1. Introduction

Sagittarius A* (Sgr A*), the $\sim 4 \times 10^6 M_{\odot}$ supermassive black hole at the Milky Way's center, has traditionally been viewed as hostile to star formation due to strong tidal shear, ionizing radiation, and dynamical heating. Yet recent multi-wavelength observations reveal the opposite: active star formation persists within a few parsecs of Sgr A*.

ALMA has detected protostellar jets and dense gas cocoons near Sgr A* (e.g., SiO outflows indicating low-mass protostars just ~ 3 light-years away). JWST reveals starbursts in Sagittarius B2, contributing disproportionately to galactic star formation. GRAVITY observes dozens of young stellar objects (YSOs) in the S-cluster on keplerian orbits around Sgr A*. These findings contradict naive tidal-disruption models and suggest Sgr A* may actively modulate or even seed star formation.

We propose that Sgr A* functions as a **metastable entropy modulator** within the Universal Threshold Adaptive Criticality (UTAC) framework. Rather than purely destructive, the black hole creates localized entropy minima through gravitational compression, shock triggering, and positive feedback (jets/outflows), enabling star formation in otherwise

inhospitable conditions. This is modeled as a resonant return process: gas clumps cross adaptive thresholds Θ (modulated by local density and $\beta \approx 4.8$), triggering emergent collapse and feedback loops.

2. Theoretical Framework: UTAC and Entropy Modulation

UTAC models emergent transitions via logistic functions $\psi(R) = 1 / (1 + e^{-\beta(R-\Theta)})$, where R is the control parameter (e.g., density, shear), Θ is the adaptive threshold, and β quantifies transition steepness (universal $\beta \approx 4.2-7.8$). The Frame Principle states dimensions emerge to prevent informational collapse; metastability is maintained by an entropy offset $\sigma_\Phi \approx 0.0625$.

Near Sgr A*, R includes gravitational potential gradients and feedback from accretion/outflows. When R exceeds Θ (modulated by local conditions), gas clumps collapse, forming protostars and launching jets — a resonant return of entropy and mass-energy. This is thermodynamically conservative and observationally testable.

3. Observational Anchors

- **ALMA**: Protostellar jets and SiO outflows within ~ 3 ly of Sgr A* indicate low-mass star formation despite tidal shear (Yusef-Zadeh et al., 2025; ALMA Cycle 10).
- **JWST**: Starbursts in Sagittarius B2 and dense cores with mature stellar populations (Schuller et al., 2025; JWST Cycle 2).
- **GRAVITY/VLTI**: S-cluster YSOs on keplerian orbits around Sgr A* (Gillessen et al., 2025; GRAVITY Collaboration). These observations suggest positive feedback (compression, shocks) dominates over disruption in localized regions.

4. UTAC Model for Sgr A* Star Formation

We model Sgr A* as a phase-critical node:

- $R = f(\rho, \text{shear}, \text{feedback})$
- $\Theta = \Theta_0 + \Delta\Theta(\text{local entropy, magnetic fields})$
- $\beta > 7$ in active clumps $\rightarrow$ sharp collapse and jet launch.

Feedback loops (jets compressing nearby gas $\rightarrow$ new clumps) create resonant return, maintaining $\sigma_\Phi \approx 0.0625$. This is consistent with $v_{\text{RIG}} \approx 1.352$ km/s as characteristic velocity scale for galactic dynamics.

5. Falsifiable Predictions

- $\beta > 7$ in protostellar regions near Sgr A* (testable via ALMA velocity dispersion fits).
- Asymmetric outflows distinguishable from pure tidal stripping (Gaia DR4 + ALMA).
- High-energy cosmic ray emission from cluster-driven shocks (Fermi-LAT cross-check).

6. Implementation in NeuroProfile v12

The model is integrated as a **Resonant Entropy Bridge** in NeuroProfile v12 (code/sgr_a_resonant_bridge.py). It computes σ_Φ proxies and β -fits from density/velocity data, with null-model comparison ($\Delta AIC \geq 10$) and bootstrap CIs in data/results.json.

7. Discussion & Outlook

Sgr A* demonstrates UTAC's power: black holes modulate entropy, enabling emergence rather than destruction. Future work: simulate clumps with resonant_entropy.py (Repo v12), cross-check with ALMA/JWST Cycle 3 data, and extend to other galactic nuclei.

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References (selected):

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